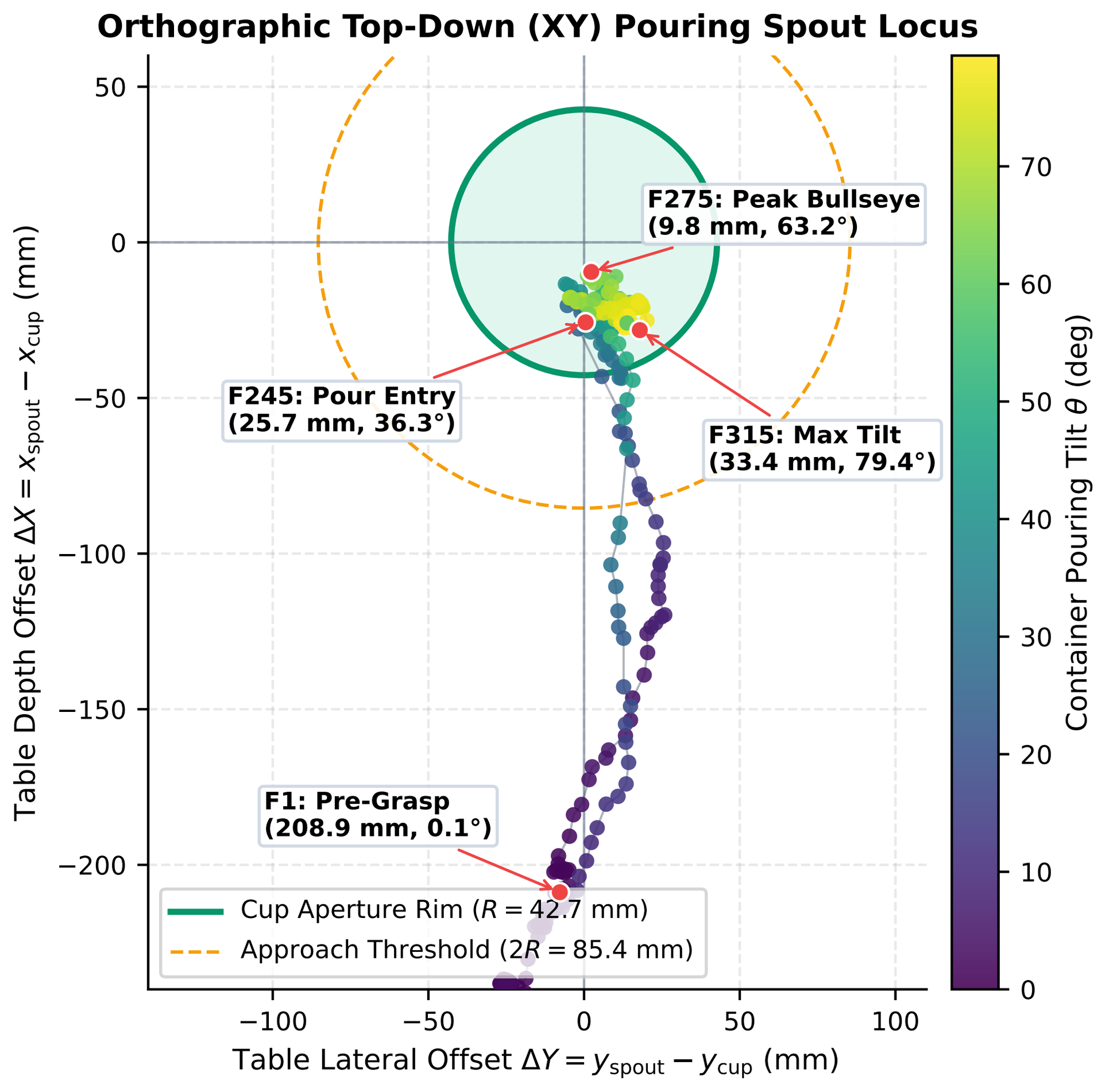
Orthographic Top-Down (XY) Pouring Spout Locus
F275: Peak Bullseye
(9.8 mm, 63.2°)
F245: Pour Entry
(25.7 mm, 36.3°)
F315: Max Tilt
(33.4 mm, 79.4°)
F1: Pre-Grasp
(208.9 mm, 0.1°)
Cup Aperture Rim ($R = 42.7$ mm)
Approach Threshold ($2R = 85.4$ mm)
Table Depth Offset $\Delta X = x_{spout} - x_{cup}$ (mm)
Table Lateral Offset $\Delta Y = y_{spout} - y_{cup}$ (mm)
Container Pouring Tilt θ (deg)
50
0
−50
−100
−150
−200
−100
−50
0
50
100
70
60
50
40
30
20
10
0